

# Email Threat Forensic Report

AI-Powered Threat Detection, GeoLocation & Forensic Intelligence Platform

## 1. Case Information

|                                     |                |
|-------------------------------------|----------------|
| Case ID: ST-1024                    | Status: Open   |
| Generated Date: 2026-09-23 16:37:11 | Priority: High |

## 2. Email Information

|                 |                                                                           |
|-----------------|---------------------------------------------------------------------------|
| Sender (From):  | PayPal Security Alert                                                     |
| Reply-To:       | support-collector@hackerserver.info                                       |
| Recipient (To): | security-target@company.org                                               |
| Subject:        | URGENT: Your PayPal Account Security Notice - Action Required Immediately |
| Date/Time:      | Wed, 23 Sep 2026 09:12:00 +0000                                           |
| Message ID:     | <202609230912.paypalsecurity@mail-auth-verify.xyz>                        |

## 3. AI Threat Analysis

| Threat Classification | Confidence | Severity |
|-----------------------|------------|----------|
| PHISHING              | High (96%) | HIGH     |

## 4. Threat Risk Score

Calculated Threat Risk Score: **92 / 100** (Severity Level: HIGH)

## 5. AI Threat Explanation

| Indicator Flagged                | Forensic Analysis & Reason                                                                        |
|----------------------------------|---------------------------------------------------------------------------------------------------|
| ✓ Sender Domain Mismatch         | Claimed organization (PayPal) does not match actual sending domain (mail-auth-verify.xyz).        |
| ✓ SPF Authentication Failed      | SPF verification failed. The IP 185.220.101.5 is not authorized by paypal.com.                    |
| ✓ DKIM Verification Failed       | Digital signature verification failed or was tampered with.                                       |
| ✓ Suspicious URL Target          | Embedded URL points to suspicious lookalike domain (paypal-verification-security-portal.com.xyz). |
| ✓ Reply-To Address Divergence    | Reply-To address (support-collector@hackerserver.info) differs suspicious from sender address.    |
| ✓ High Urgency & Threat Language | Email contains strong pressure tactics ('permanent restriction within 24 hours').                 |

## 6. Authentication Protocol Verification (SPF / DKIM / DMARC)

| SPF Status | DKIM Status | DMARC Status |
|------------|-------------|--------------|
| FAIL       | FAIL        | FAIL         |

## 7. Extracted IP Addresses

| Role                    | IP Address    | Type                    |
|-------------------------|---------------|-------------------------|
| Originating Mail Server | 185.220.101.5 | Public / External Relay |

## 8. GeoLocation & Threat Infrastructure

|            |               |
|------------|---------------|
| Source IP: | 185.220.101.5 |
| Country:   | Germany (DE)  |

|                |                                     |
|----------------|-------------------------------------|
| City / Region: | Munich, Bavaria                     |
| ISP:           | Tor Exit Node Network / Host Europe |
| ASN:           | AS208323                            |
| Reputation:    | High Risk (Tor / Malicious Relay)   |
| Data Source:   | DEMO / SAMPLE GEOLOCATION CACHE     |

Note: IP geolocation represents approximate network infrastructure location and does not identify the attacker's exact physical location.

9. Extracted Domains and URLs

No embedded HTTP/HTTPS URLs extracted from message body.

10. Threat Intelligence Correlation

|                                                  |                                               |
|--------------------------------------------------|-----------------------------------------------|
| Query IOC: 185.220.101.5                         | Reputation: MALICIOUS                         |
| Categories: Tor Proxy, Phishing Host, Spam Relay | Blacklist Detection: 14 / 88 security vendors |

11. Forensic Timeline (Header Hop & Investigation Flow)

| Step / Hop | Server Host / Action                                        | IP Address    | Timestamp / Details             |
|------------|-------------------------------------------------------------|---------------|---------------------------------|
| Hop #1     | mail-auth-verify.xyz (mail-auth-verify.xyz [185.220.101.5]) | 185.220.101.5 | Wed, 23 Sep 2026 09:12:05 +0000 |

Investigation Workflow

Email received → Header extracted → Indicators identified → AI analysis completed → Threat detected → Alert generated → Forensic case created

12. Evidence Summary & Hashes

| Artifact Type                   | Value / Descriptor                                               |
|---------------------------------|------------------------------------------------------------------|
| Sender Domain                   | mail-auth-verify.xyz                                             |
| Header Hash (MD5 Demo)          | e99a18c428cb38d5f260853678922e03                                 |
| Body Payload Hash (SHA256 Demo) | a591a6d40bf420404a011733cfb7b190d62c65bf0bcda32b57b277d9ad9f146e |

13. Recommended Security Action

- 1. Block sender domain and originating server IP across gateway firewalls.
- 2. Do NOT interact with embedded links or download attachments.
- 3. Reset corporate credentials immediately if user clicked embedded URLs.
- 4. Dispatch threat indicator IOCs to SIEM / EDR endpoints.